

Koi 8×8: A low-cost 64-channel current-source-and-sense controller for thermo-optic phase shifters in photonic integrated circuits

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Abstract

This article describes the development of *Koi 8×8*, a compact, low-cost, open-source 64-channel current-source-and-sense controller for driving arrays of thermo-optic (TiN) phase shifters on photonic integrated circuits. The system is built entirely from off-the-shelf parts: a Raspberry Pi Pico controls up to eight stackable daughterboards, each combining an 8-channel DAC, a 24-bit $\Sigma\Delta$ ADC, and eight precision industrial current transmitters operated as current sources. Each channel sources a programmed current and reads back the load voltage, so the host computes heater resistance and dissipated power in situ for closed-loop tuning. On each board the DAC and ADC share one precision reference, so the reference error cancels in the derived resistance; a one-time calibration against a reference source-measure unit recovers absolute accuracy. Complete hardware, firmware, host software, and calibration tooling are released under open licenses.

Specifications table

Hardware name	Koi 8×8 — 64-channel current-source-and-sense controller
Subject area	• Engineering and materials science • Optics and photonics
Hardware type	• Electrical engineering and computer science • Measurement / instrumentation and control
Closest commercial analog	Multichannel source-measure systems for photonic biasing, e.g. Qontrol Q8iv/BP modules and multi-channel Keithley 2400-class SMUs.
Open source license	Hardware: CERN-OHL-S v2; Firmware/software: MIT
Cost of hardware	
Source file repository	(mirror:)

1 Hardware in context

Programmable photonic integrated circuits (PICs) have become a central platform for optical communications, sensing, microwave photonics, and increasingly for optical computing and quantum information processing [1]. What makes these circuits *programmable* is a dense fabric of tunable elements — meshes of Mach–Zehnder interferometers and ring resonators whose phase is set

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electrically [1, 2]. The dominant tuning mechanism is the thermo-optic effect: a resistive micro-heater, commonly a titanium-nitride (TiN) filament patterned over the waveguide, dissipates electrical power and locally shifts the refractive index [3]. A single modern device can integrate from tens to several hundred such heaters, and each must be held at an independent, stable bias for the circuit to hold a programmed state.

Two requirements make driving these heaters non-trivial at scale. First, the relevant control variable is dissipated *power*, not drive current, and the heater resistance drifts with temperature and ages over time; robust operation therefore benefits from reading back the voltage across each heater so that resistance and power can be computed and servoed in situ, rather than driven open-loop [4]. Second, the channel count is large. Driving every heater from a laboratory source-measure unit (SMU) is accurate but prohibitively expensive and bulky beyond a handful of channels. Dedicated solutions exist at both ends of the spectrum: commercial multichannel driver modules (e.g. Qontrol [5, 6]) offer dense drive with optional readback but at a substantial per-channel cost, while research-grade custom CMOS controllers [7] demonstrate excellent density and power but require ASIC fabrication and are not reproducible open hardware. On the open-source side, source-measure designs are typically single-channel bench instruments (e.g. the uSMU [8]); to our knowledge no published open design combines dense multichannel current drive with per-channel voltage readback at the scale photonic meshes need. There remains a need for a low-cost, fully open, and readily reproducible controller that combines dense current drive with per-channel readback.

This paper presents *Koi* 8×8 , an open-source controller named for its stack topology — eight stackable daughterboards of eight channels each, not any particular photonic mesh geometry — that drives up to **64 independent current-source channels** and measures the voltage developed across each load, so the host can compute per-channel resistance and power in situ. The design targets the accuracy regime that matters for heater tuning: *relative / ratiometric* accuracy. Each daughterboard carries its own precision voltage reference (eight in total), shared by that board’s DAC and ADC, so the commanded drive and the measured readback are scaled by the same reference and its error cancels in the derived resistance; channels on the same board are additionally compared against the same reference, so common-mode errors cancel between adjacent tuning elements. Board-to-board matching is bounded by the reference initial tolerance and is characterized in Section 7. Absolute accuracy is available through an optional one-time calibration against a reference SMU.

Contribution.

- A stackable controller + daughterboard architecture scaling from 8 to 64 channels from commodity parts (RP2040, DAC80508, AD7193, XTR200).
- Per-channel closed-loop readout enabling in-situ R and power measurement, with an analytic correction for the on-board sense divider current.
- A fast continuous-conversion scan that reads a two-board, 16-channel system in ≈ 80 ms (measured), $\sim 40\times$ faster than per-channel single-shot conversion.
- Open hardware, firmware, host software, and calibration tooling.

2 Hardware description

The system is a host-commanded instrument. A PC sends per-channel current setpoints (in mA) over USB serial; the firmware sources the commanded current through each load and reports the measured voltage, from which the host computes resistance and power.

2.1 Signal chain (per channel)

Each of the 64 channels is an independent loop: an 8-channel DAC sets the input of a precision industrial current transmitter (Texas Instruments XTR200, a 3-wire current/voltage transmitter)

operated as a current source, the sourced current flows through the external load (a TiN heater in the target application) to ground, and an 8-channel 24-bit $\Sigma\Delta$ ADC measures the voltage developed across the load through a 6:1 input divider.

Using an industrial transmitter outside its headline 4–20 mA niche is a deliberate design choice, and it is what makes the channel density affordable: the XTR200 integrates the complete precision output stage in one commodity package ($\sim$ US\$1.5), absorbs the sourcing dissipation in a part designed for continuous industrial drive, and its datasheet figures — span error $\leq \pm 0.065\%$, nonlinearity $\pm 0.003\%$, output impedance $47\text{ G}\Omega$ — match or beat a discrete op-amp current source that would cost more board area, calibration effort, and thermal management per channel. Eight sources per daughterboard therefore add no exotic parts and no heatsinking. The design’s economy comes not from a new circuit but from recognizing that a mass-manufactured industrial part — cheap because it is produced in volume for a market far larger than laboratory instrumentation — already integrates the entire precision source channel; the contribution is the system this enables.

The XTR200’s internals are themselves the de-facto standard current-transmitter architecture — a two-op-amp, sense-resistor-feedback voltage-to-current converter: an input amplifier servos the control voltage across a precision set resistor, and a second amplifier with an integrated PMOS pass transistor reproduces the resulting current at the output. An early revision of this design implemented the same topology discretely with a precision dual op amp (OPA2192) per channel. The integrated part displaced it on three grounds that compound at 64 channels: (i) its transfer function is set by internally trimmed thin-film resistors, so the transconductance is uniform across channels and calibration collapses to a single per-channel offset-current reading (Section 7), whereas a discrete channel’s gain inherits the tolerance of its external resistors and would require a two-parameter fit on every channel; (ii) its $47\text{ G}\Omega$ output impedance renders the source effectively ideal, leaving the sense divider’s known $120\text{ k}\Omega$ shunt as the only load-dependent error — which is what permits the analytic correction in the measurement model below; and (iii) the output stage, short-circuit protection, and active-low output-disable pin (used here for power-up sequencing of the per-board front-end enables) arrive integrated, where a discrete channel would add a pass transistor, protection network, and enable gating per channel — board area that would break the eight-channels-per-daughterboard density.

2.2 Architecture

A single RP2040 (Raspberry Pi Pico) controller drives up to eight identical daughterboards. Each daughterboard carries one **DAC80508** (8-ch, write-only), one **AD7193** (8-ch ADC), eight **XTR200** current sources, and its own **REF5030** 3.0 V precision reference shared by that board’s DAC and ADC — the basis of the per-board ratiometric accuracy argument (Section 1). Addressing uses two 74HC138 3-to-8 decoders (one per ADC/DAC bank) for chip-select, and one SN74LV595 shift register to gate each board’s analog front ends.

- **SPI0** (shared): AD7193 ADCs + the SN74LV595 front-end enable register. Mode 3, 1 MHz.
- **SPI1** (write-only): DAC80508 bus.
- **Addressing**: shared address lines select daughterboard k via the inverted decoder mapping (address = $8 - k$); one bank enable (ADC_EN / DAC_EN) is asserted at a time.

2.3 Measurement model

The reported voltage is the raw ADC-pin voltage. The true load voltage is recovered with the 6:1 sense divider: $V_{\text{heater}} = V_{\text{ADC}} \times 6$. The divider ($100\text{ k}\Omega + 20\text{ k}\Omega = 120\text{ k}\Omega$) draws a small, *known* current from the load node, corrected analytically:

$$I_{\text{heater}} = I_{\text{source}} - \frac{V_{\text{heater}}}{120\text{ k}\Omega}.$$

This correction is exact and computed from the live measurement (not a calibrated parameter). The XTR200 transconductance ($\approx 0.47 \text{ V/mA}$ seed) and a per-channel offset current (I_{OS}) are captured by the calibration in Section 7.

3 Design files summary

Design file name	File type	Open source license	Location of the file
Controller board (schematic + PCB)	KiCad / Gerber	CERN-OHL-S v2	
Daughterboard (schematic + PCB)	KiCad / Gerber	CERN-OHL-S v2	
Firmware (<code>src/</code> , <code>include/</code>)	C++ (PlatformIO)	MIT	
Host GUI (<code>koi_gui.py</code>)	Python	MIT	
Calibration / sweep tooling	Python	MIT	
Bill of materials	CSV / XLSX	CC-BY-4.0	

4 Bill of materials summary

Designator / component	Qty	Unit cost (USD)	Total cost (USD)	Supplier	Supplier Part No.	Manufacturer Part No.
Raspberry Pi Pico (RP2040) controller	1					
DAC80508 (8-ch DAC)	8					
AD7193 (8-ch 24-bit ADC)	8					
XTR200 current source	64					
74HC138 decoder	2					
SN74LV595 shift register						
REF5030 3.0 V precision voltage reference (one per daughterboard)	8					
Sense divider resistors (100 k/20 k)						
Passives, connectors, stacking headers (consolidated)	—	—			—	—
PCBs (controller + 8 daughterboards)	9					

Designator / component	Qty	Unit cost (USD)	Total cost (USD)	Supplier	Supplier Part No.	Part	Manufacturer Part No.
PCB assembly (fine-pitch reflow / PCBA service)	9				—		—
System total							

Placeholder values — table will be updated with final quantities, unit costs, and supplier/manufacturer part numbers from the released BOM spreadsheet.

5 Build instructions

```
pio run -t upload      # build + flash the RP2040 over USB
pio device monitor     # serial console @ 115200
```

6 Operation instructions

The firmware runs a line-based ASCII command server over USB serial (115200, \n-terminated). Each command returns exactly one reply line. Channels use a flat global index $g = \text{board} \times 8 + \text{channel}$ (0–63); unpopulated channels report **nan**. Populated boards are auto-detected at boot. Every OK reply echoes its command name so the host can confirm a reply belongs to the command it sent, and every firmware-initiated line begins with # (boot banner, calibration progress, **STREAM** dumps, diagnostics), so an unprefixed line is always a direct command reply.

Command	Effect	Reply
*IDN?	identify	KOI,8x8,fw1.1
*RST	all currents 0, front-ends disabled	OK *RST
ISETA i0 ...i63	set all 64 currents (mA, g -order)	OK ISETA
ISET g mA	set one channel	OK ISET <g>
MEASA? [mask]	measure all 64 (or masked boards)	v0,...,v63
MEAS? g	measure one channel	<volts>
XTR enmask	bit $b = 1 \rightarrow$ board b front-ends enabled	OK XTR 0x..
AVG n	on-micro averages per measurement (1–64)	OK AVG <n>
RATE fs	AD7193 filter word (speed/noise)	OK RATE <fs>
STREAM ON OFF	periodic # dump for monitoring	OK STREAM ON OFF
RESCAN	re-detect boards at runtime	OK RESCAN active=0x.. new=0x..
PING? b	board liveness (ADC ID + one conversion)	OK PING 1 0

A typical experiment loop is **ISETA ...** (set), settle, **MEASA?** (read). A Tkinter host GUI (`koi_gui.py`) polls **MEASA?** and displays raw and divider-corrected voltage for all 64 channels in an 8×8 grid; `sweep_ch0.py` performs a scripted single-channel current sweep.

7 Validation and characterization

7.1 Calibration method

Bench setup. Every channel is terminated in a known load resistor standing in for the TiN heater: a thin-film $1\text{ k}\Omega$ part with $\leq 25\text{--}50\text{ ppm/K}$ temperature coefficient in a large package (to bound self-heating: 36 mW at 6 mA), whose value is taken from an in-situ four-wire measurement rather than from the part grade. A dedicated calibration jig PCB carries all 64 loads, permanently connected so that every channel remains loaded and driven throughout the campaign. Each load carries Kelvin taps at its resistor pads — a sense pair and a force pair — and a 23-channel bank of four-pole relay stages (paired DPDT signal relays) connects one load’s taps at a time to a 6.5-digit DMM (Keithley 2100), operated in offset-compensated four-wire mode for resistance scans so that any constant residual current through the load cancels; a re-positionable sense-only harness maps the relay bank onto each third of the array in turn, without ever interrupting a drive path. The same taps thus serve two functions: four-wire resistance measurement of every load through the jig itself (repeatable at any time, e.g. warm, to bound drift), and Kelvin voltage sensing during the current sweeps, so every sweep point yields the load voltage directly and the channel current as

$$I = \frac{V_{\text{DMM}}}{R_{\text{meas}}} + \frac{V_{\text{DMM}}}{120\text{ k}\Omega},$$

the second term being the known sense-divider current (Section 2). Because the DMM is switched across the loads in parallel — never inserted in series — the current path is never broken: all 64 channels remain loaded and driven throughout the scan, i.e. the device is characterized under representative operating conditions rather than one channel at a time into an otherwise idle board. Mechanical relays are used (open-contact leakage in the pA range) so the jig itself cannot contribute at the microampere level being resolved. A metal-foil transfer-standard resistor and a Keithley 2400 source-measure unit inserted in series on a small number of channels provide an independent cross-check of the voltage-over-resistance method. The jig design files (schematic, PCB, control script) are released with the paper.

Sweep protocol and fit. Each channel is swept over $0.25\text{--}6\text{ mA}$ in both directions — the linear operating regime; sub- $100\text{ }\mu\text{A}$ points are excluded because the source cannot sink current, so readings near zero bottom out rather than extrapolate linearly. A per-channel linear fit of commanded versus measured current yields `calSlope` (the transconductance, expected $\approx 0.47\text{ V/mA}$ and shared across channels, consistent with the trimmed-thin-film argument of Section 2) and `calOffset` (the per-part offset current I_{OS} , approximately equal to the enabled zero-command reading). The calibration proper therefore uses *only* the $1\text{ k}\Omega$ loads; many sweep points serve to verify linearity and average noise, not to fit additional parameters.

Separating current-side from voltage-side errors. A single-load sweep, however many points it contains, carries a structural confound: with a fixed load, $V = IR$ is perfectly correlated with I , so an error term that scales with *current* (XTR200 gain error, true I_{OS} , DAC error) is indistinguishable from one that scales with *voltage* (board leakage, the $V/120\text{ k}\Omega$ divider correction, ADC or divider gain error) — both produce identical residuals. To break the correlation, one board is re-run with its loads exchanged for $\sim 300\text{ }\Omega$ and $\sim 2\text{ k}\Omega$ parts of the same grade: at the same 3 mA setpoint the load then sees ~ 0.9 , ~ 3 , and $\sim 6\text{ V}$ across the three load sets — the same current at a $\sim 7\times$ span of voltage. Comparing the apparent per-channel offset across the three loads attributes it: an offset that is invariant with load is genuinely current-side (real I_{OS} , and belongs in `calOffset`), whereas one that scales with V is voltage-side (leakage or divider-correction error) and must *not* be folded into `calOffset` — doing so would be correct only at the calibration load and would mis-correct on a real heater of different resistance. The same re-run validates the analytic divider correction directly, exercises a wider portion of the ADC

span, and yields a load-independence figure for the source. This is an attribution measurement, not a three-point calibration: if it confirms the error budget is current-side, the remaining seven boards need only the 1 k Ω dataset.

7.2 Current-mode characterization

The current-drive characterization mirrors the per-parameter methodology of a representative commercial module (Qontrol Q8a [5]) so the two can be compared plot-for-plot; all figures below report *Koi*'s own measured data. Note the differing operating ranges (Koi 0–6 mA vs. Q8a 0–100 mA), so resolution/precision are additionally reported as a fraction of full scale. *Preliminary bench data (single board, 2026-06-17)*: XTR200 gain is near-ideal (datasheet span error $\leq \pm 0.065\%$, nonlinearity $\pm 0.003\%$), so `calSlope` is treated as a shared constant; the per-channel offset I_{OS} is the term needing measurement, bench-measured $\approx \pm 4 \mu\text{A}$ (datasheet $\pm 10 \mu\text{A}$ max).

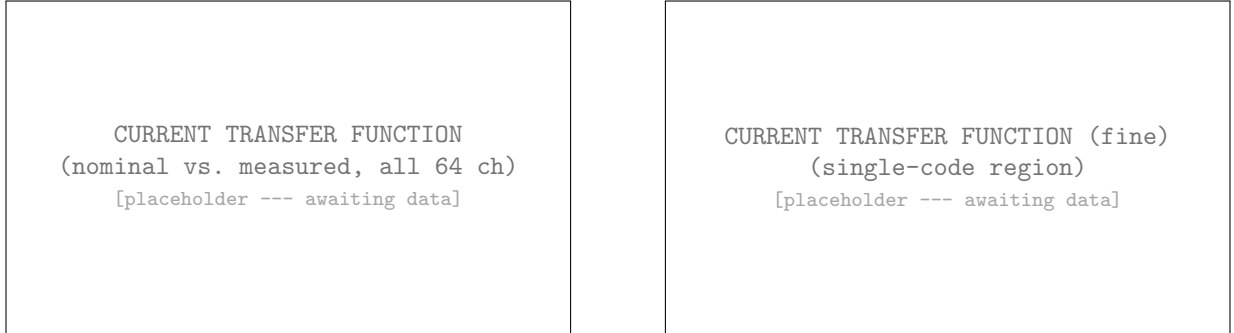


Figure 1: Current transfer function. Nominal commanded current vs. current measured by an external reference (Keithley 2400) across all 64 channels; the fine panel resolves the low-current, single-code region.



Figure 2: Current deviation from nominal. Left: against an external reference DMM (source accuracy). Right: against the on-board AD7193 readback (closed-loop measurement accuracy, after the sense-divider correction).

7.3 Sense-divider correction

7.4 Noise and resolution

7.5 Thermal validation

Section 2 claims the design needs no heatsinking because the dissipation is distributed: worst case per transmitter is roughly $(V_{\text{supply}} - V_{\text{load}}) \times I \approx 120 \text{ mW}$ (6 mA into a near-short), i.e. $\sim 1 \text{ W}$ per fully driven board spread across eight packages — the same distributed-dissipation approach used by commercial multichannel drivers, so this is standard practice to be *validated*, not a novelty to be claimed.

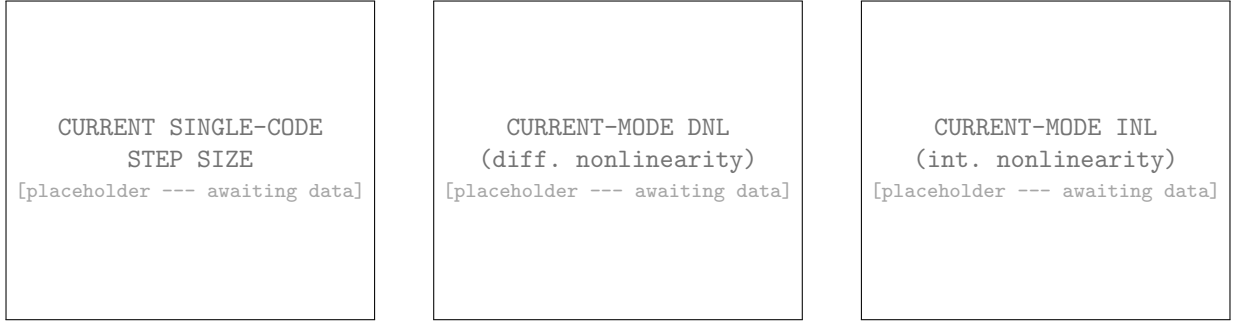


Figure 3: Code-level linearity. Single-code current step-size distribution (nominal LSB set by the DAC80508 over the 0–6 mA span), and current-mode differential (DNL) and integral (INL) nonlinearity.

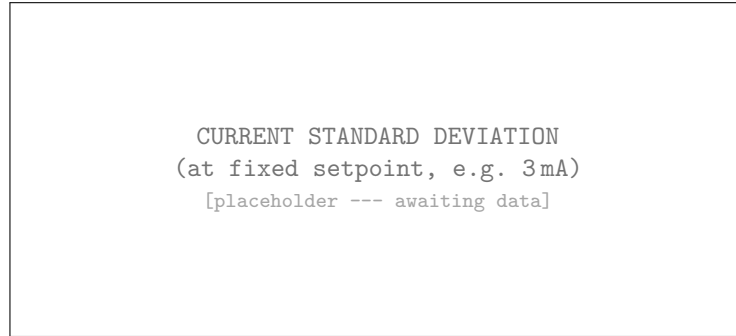


Figure 4: Current precision: distribution of repeated measurements at a fixed commanded current, quantifying single-channel noise (reported as 1σ).

7.6 Throughput

The continuous-conversion sequencer reads one full 8-channel board per pass instead of per-channel single conversions, reducing a two-board, 16-channel readout from ≈ 3.4 s to ≈ 80 ms (measured) — gated by the filter rate rather than a fixed per-channel restart penalty.

7.7 Channel-to-channel crosstalk and board reproducibility

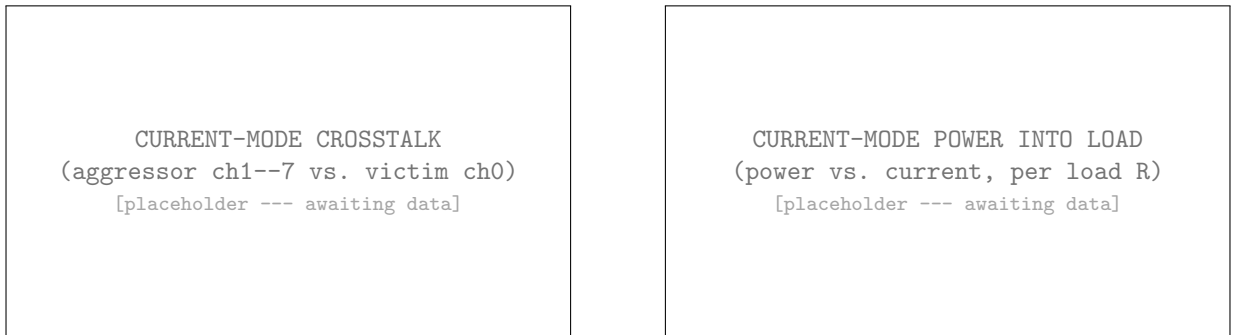


Figure 5: Left: current-mode crosstalk — shift in a victim channel’s current as the other seven channels are swept across their range. Right: power delivered to representative loads vs. commanded current, with the nominal maximum-power point marked.

8 Capabilities and use cases

Declarations

References

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